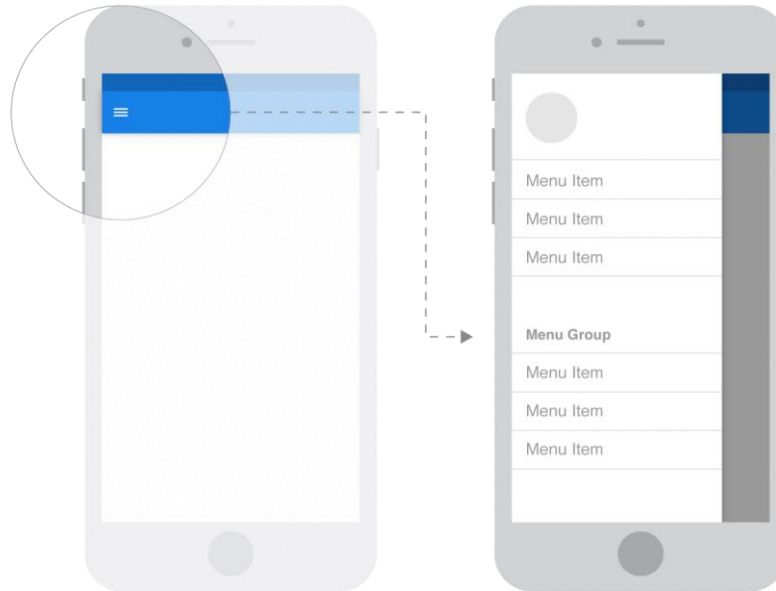


Mobile UI

Diogo Cabral
diogo.cabral@isel.pt

Mobile User Interface Patterns

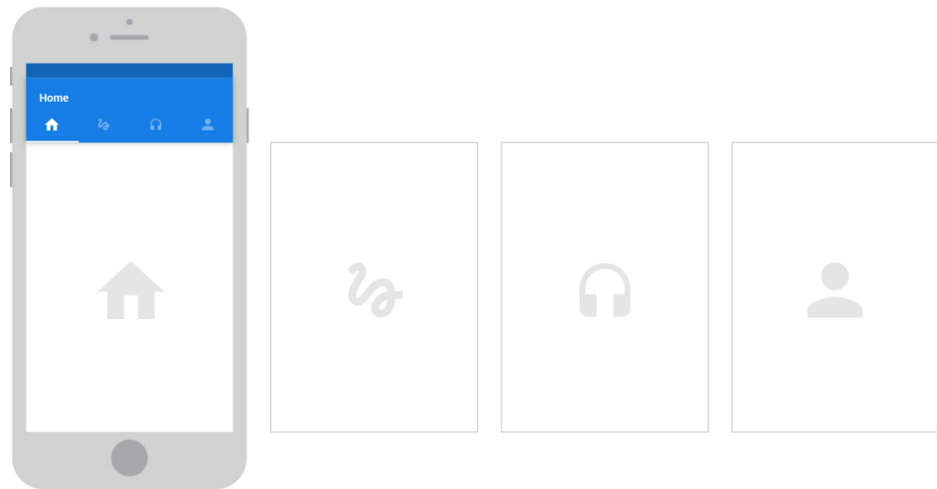
Navigation: Sliding Drawer (Hamburger Menu)



- Android or in mobile Web browser based apps
- Allows to have an indefinite number of elements without cluttering the initial experience
- Suitable to applications with many sections with equal importance
- Higher number of entries and submenus are the major problems

Navigation: Upper Area Tabs

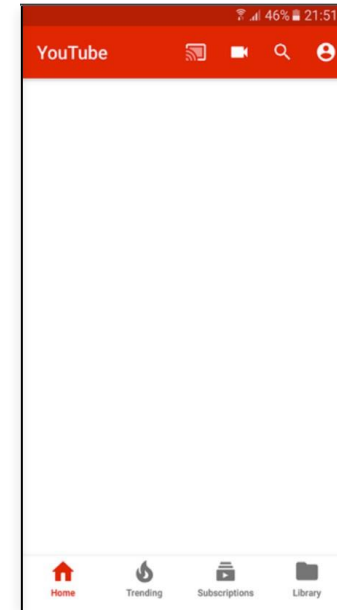
Web apps and native applications, being more common in Android applications



- Not good for many menu options
- Useful for users not accustomed to the mobile interfaces
- Can coexist alongside hamburger menus

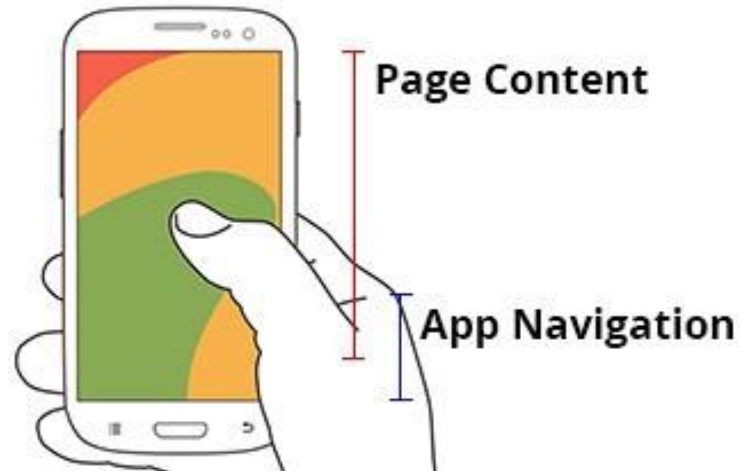
Navigation: Bottom Menu

Current trend in Android and iOS



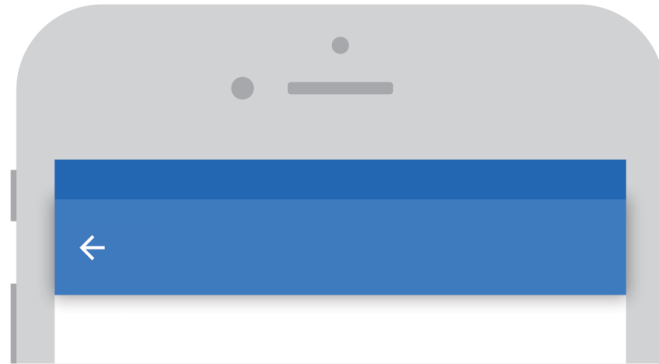
- Bottom menus are close to the user's thumb (easy to change with one hand)
- Dividing the content of the application into a maximum of five sections (all the sections always visible and with enough touchable space)
- Needs other navigation techniques for more than five sections

Navigation: Action in the Screen



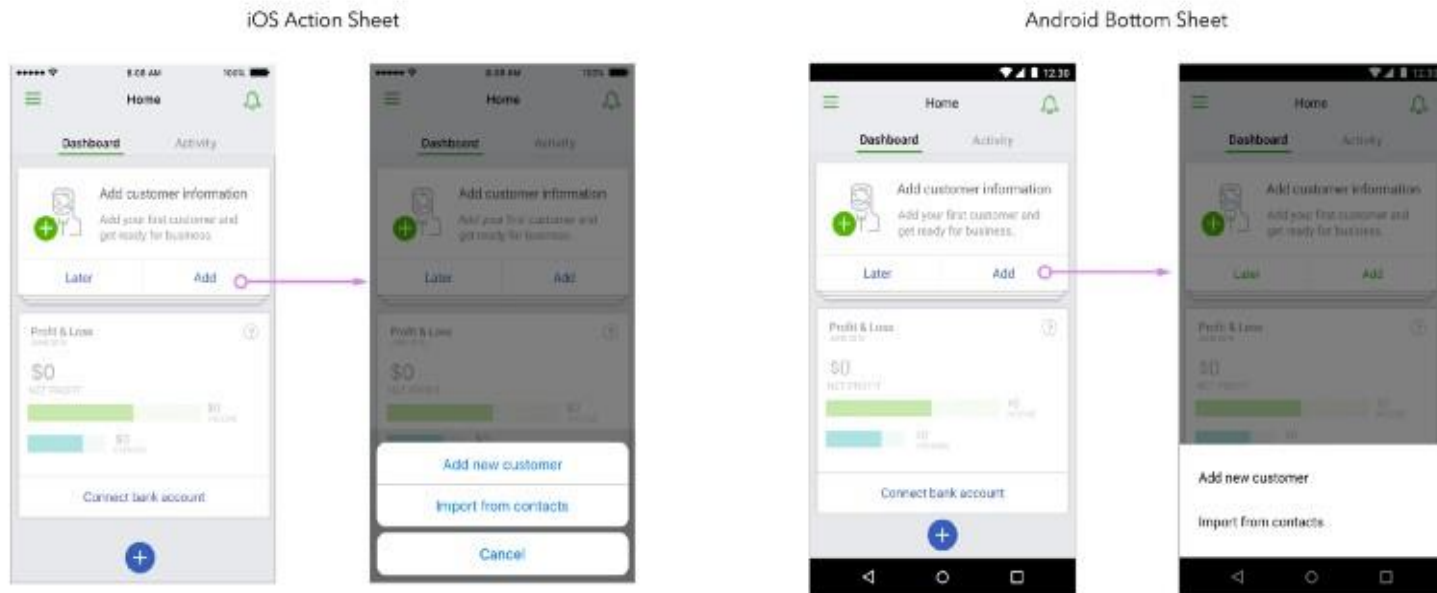
- Mobile screens are generally getting bigger over the recent years
- Important to locate the most often used actions near the thumb area
- Steven Hoober discovered that 49% of the people use the mobile with one hand

Navigation: Back Button



- Used in complex processes with several steps
- Used when a user visits a detail of any item found in a previous screen
- Placed on the top-left (as in desktop browsers)
- Languages written from right to left (e.g., Arabic or Hebrew)
 - Arrow will be placed and pointing to the right-hand side

Navigation: Floating Action Buttons



- Used in applications to highlight a functionality over the other elements on the screen
- Usually located in the bottom area of the screen because of its proximity to the thumb area

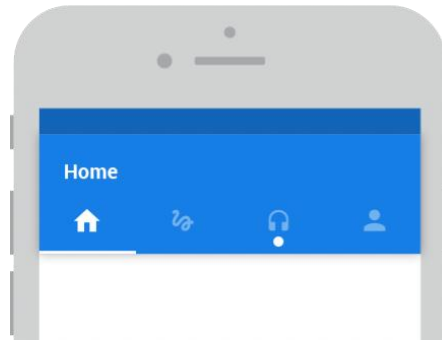
Navigation: Bottom Sheets



- For sub-flows instead of drop-downs, hamburgers side drawers or pop-up dialogues
- Facilitate vertical scroll for additional content and horizontal scroll (carousel) for similar content without jumping screens

Notifications: Circle Hint

Notifications can cause unnecessary distraction to users



- Circle indicating something has changed is fairly accepted
- It is common to find it both with a number inside
 - The number represents the number of new items

Dialogs: Careful Use



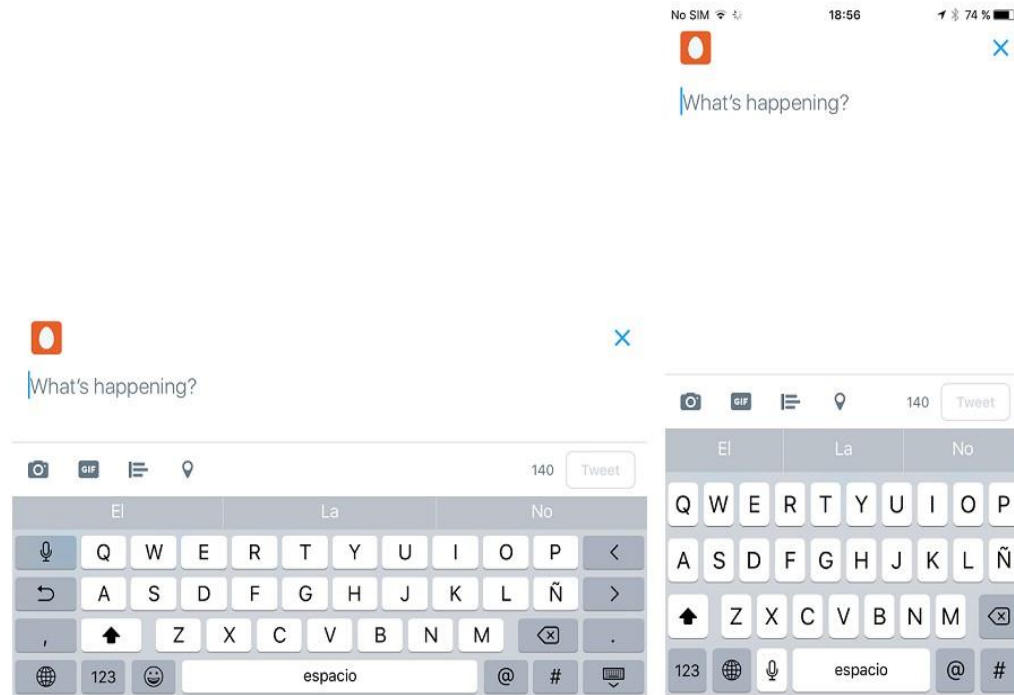
- Block any other user activity forcing users to reply immediately
 - This makes dialogs quite intrusive and annoying
- Used only when the application can not continue without user interaction
- Better to use different design options like options that appear from the bottom (bottom sheets)

Galleries with Miniatures



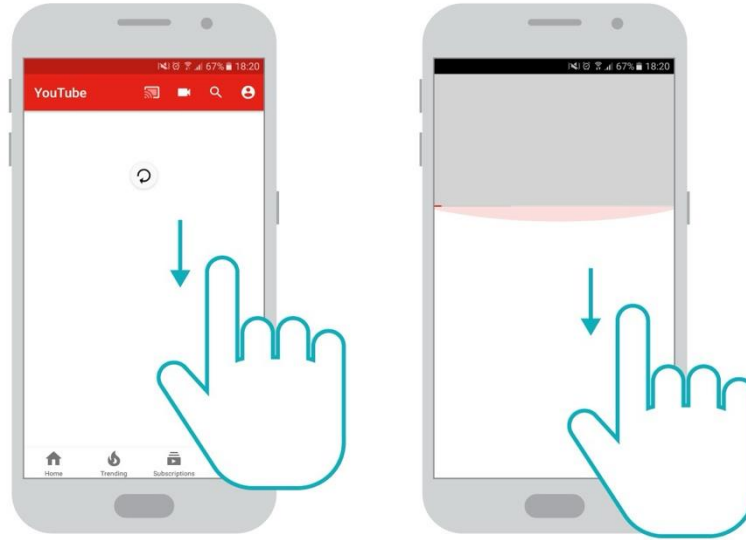
- Display lists of items in a graphical way in applications and mobile systems
- Items shown as thumbnails should be interactive

Portrait and Landscape Orientations



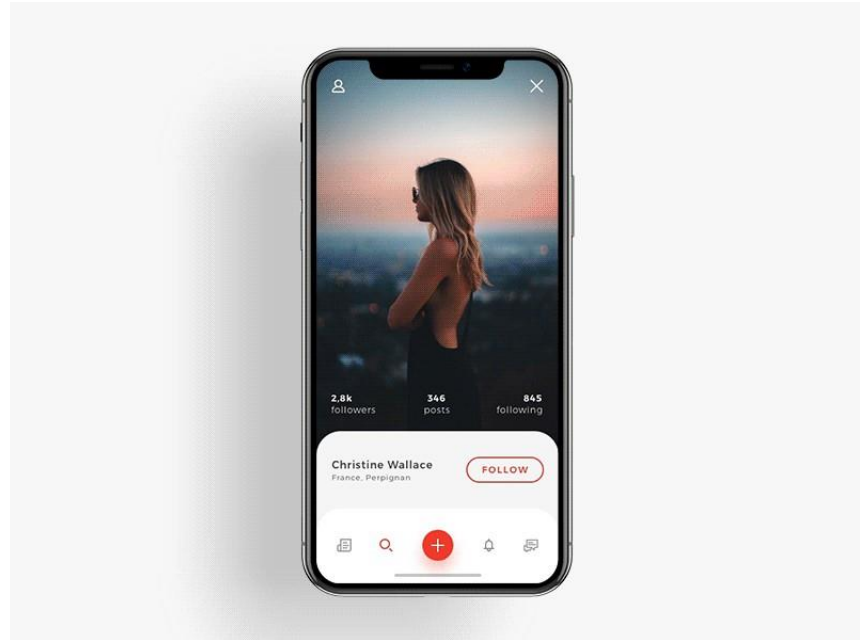
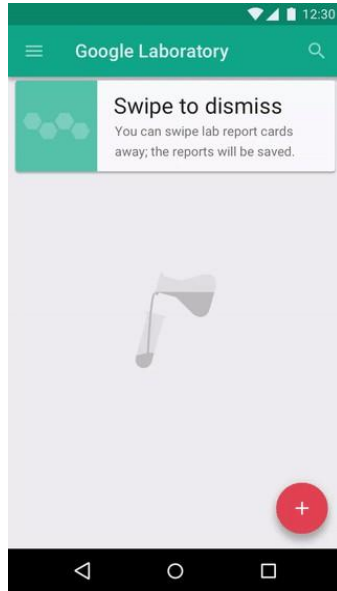
- Most users use their mobile in portrait position
- Applications may take advantage of the use of the landscape position by introducing new ways of viewing content
- Twitter show a larger keyboard in horizontal position

Swipe Down to Refresh



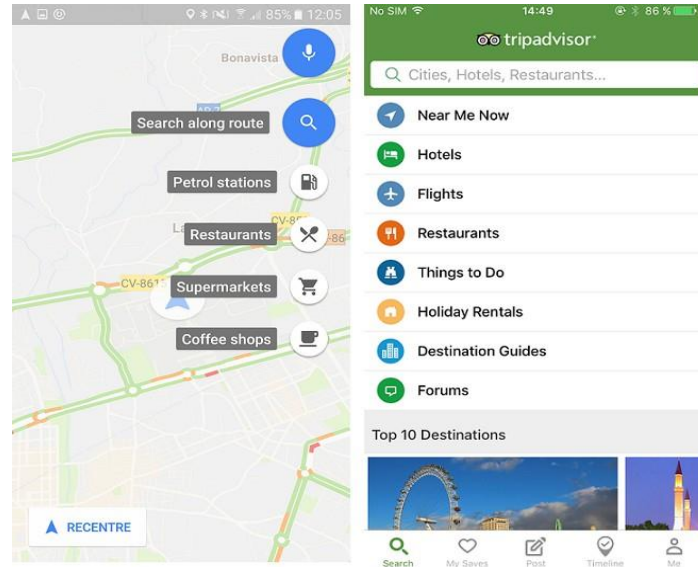
- Important to give feedback to the user
- Screens that not support this type of interaction should add clues to make the user understands

Gestures with Animations



- Animations are used to make clear what interaction are available
- Animations to clarify the spatial relationships between user interface elements
- Animations to provide feedback on the interaction

Search Suggestions



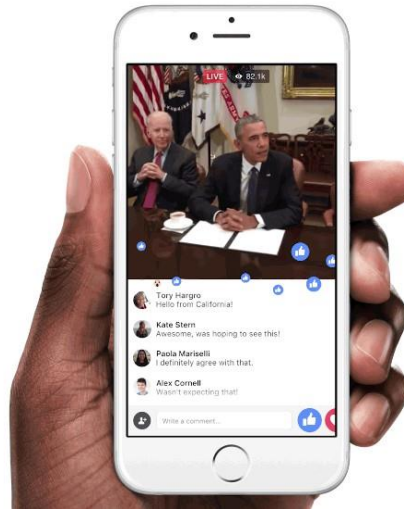
- In complex applications it is difficult to reach all content just by browsing menus
- Search boxes can help the user finding the content
- Search boxes with suggestions can help the user and reduce de user input

Full-Screen Experiences



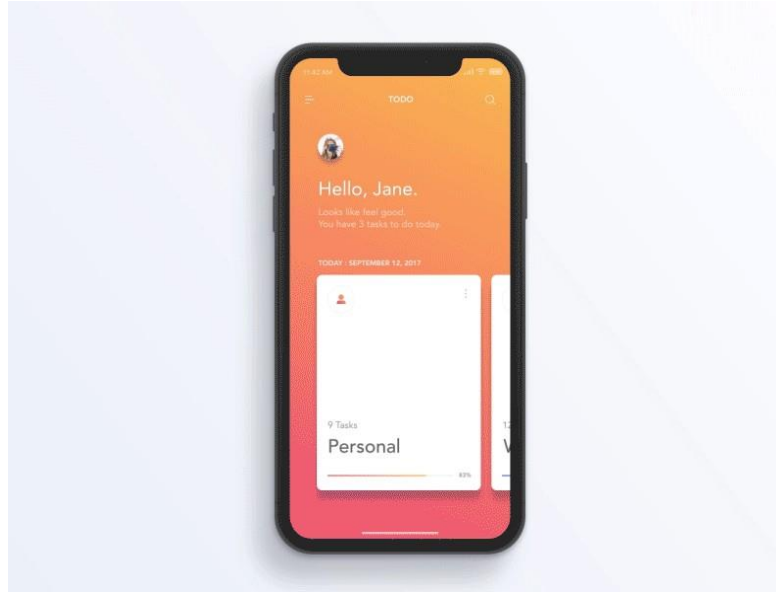
- More screen space available for users
- Images should not appear pixelated on mobile screen

Video Dominance



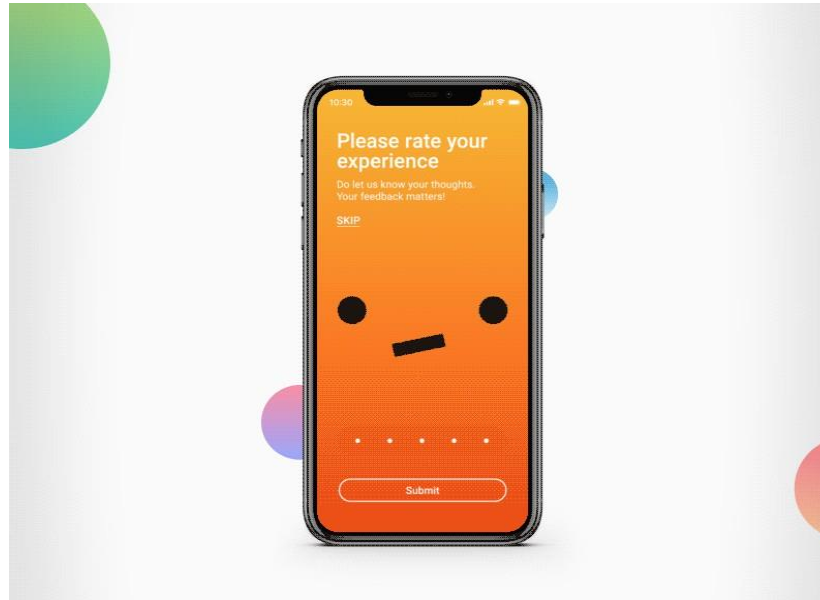
- According to Hubspot, 78% of people watch videos online every week, and 55% watch videos online every day
- Adapting Video for Portrait Orientation (94% of the time the mobile is used in portrait orientation)
- The average human attention span has fallen from 12 seconds in 2000, to 8 seconds in 2015. Video presentation should be adapted to this change

Color As Functional Element



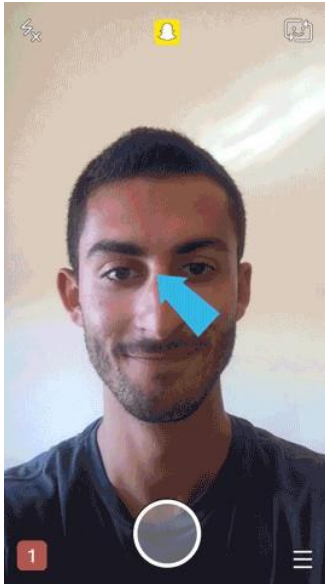
- In mobile apps, color is not only for aesthetics but it is also a part of the functional experience
- For instance, use color to visually separate different types of notifications

Mimic Emotions In UI Feedback

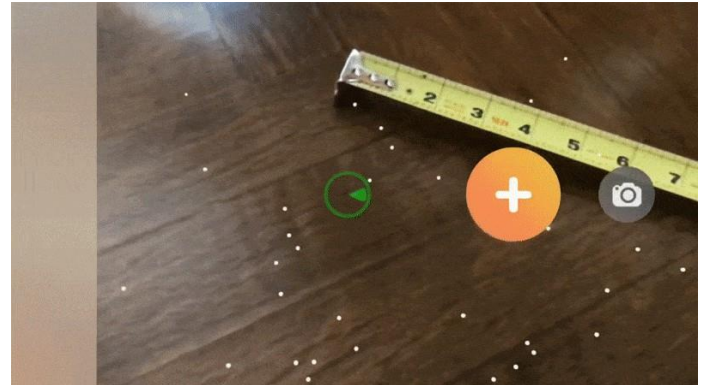


- Humans establish an emotional connection with all products they use
- Humans expect some level of human-like feedback when they interact with products
- Even though humans know that the products are not humans and can not feel emotions, users want to believe that they can

Augmented Reality



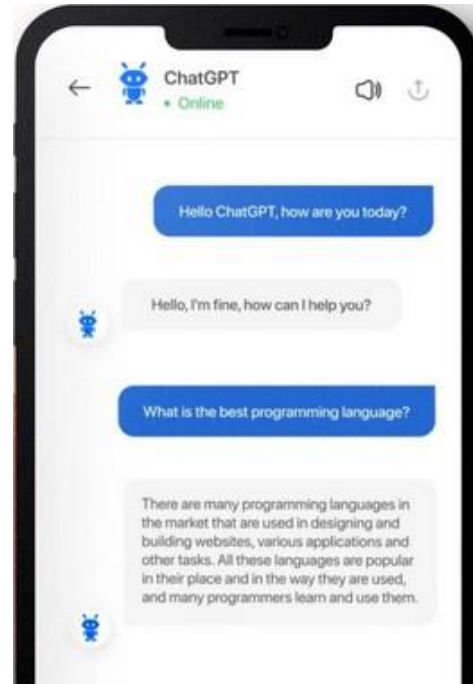
Entertainment Applications



Real-life Problems

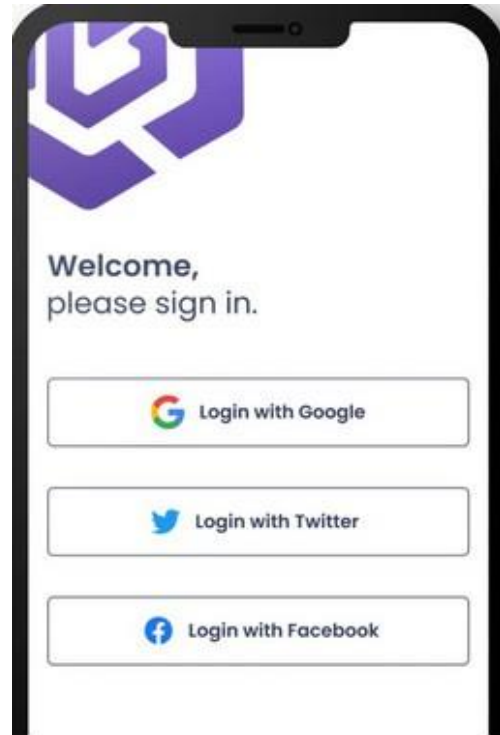
Mobile User Interface Patterns Trends

The Dominance of Chatbots



- Redefined the personalisation properties of any business
- It also has an increased responsive experience for users
- Can easily comprehend the intent of the user (using AI) and present to them the exact and relevant information they seek

Passwordless Login



- Improving the UX
- Remembering multiple passwords is a problem
- Login secure using Biometric or phone based authentication

Flat Design



- Devoid of any gradients or shadows (Black or White background)
- Can prioritise content instead of visual clutter
- Minimal textures and simple shapes, flat design ensures that responsive designs load smoothly and fast improving UX

Minimalism



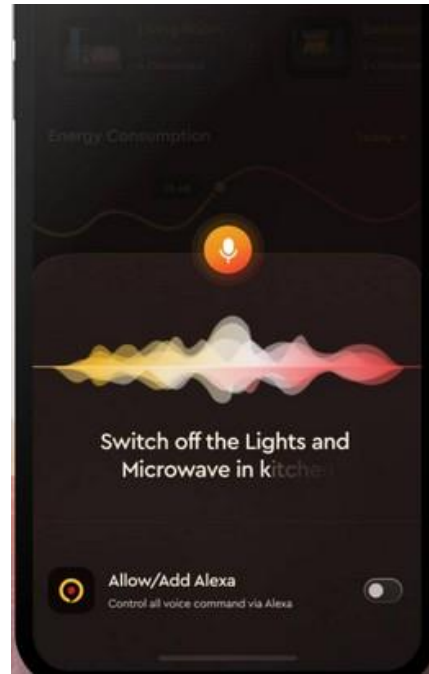
- Emphasises on clarity, simplicity, and functionality
- Prioritises content over visual noise and focuses on the app's functionality
- Minimalism can easily adapt to various resolutions and screen sizes, making it appropriate for responsive design

Asymmetrical Elements



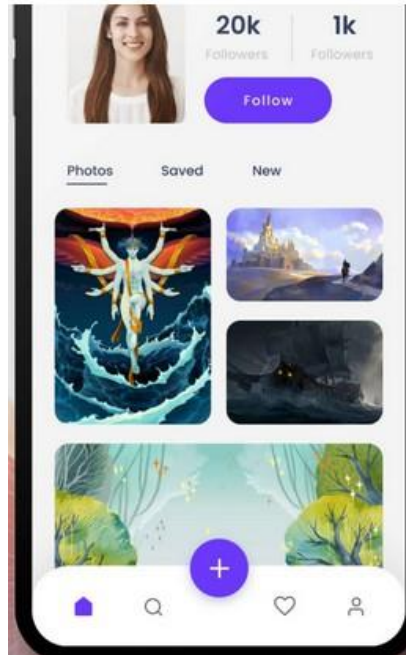
- Visual elements intentionally arranged irregularly
- Can assess an asymmetrical design by checking line weight changes, irregular sizing, contrasting colours, and different shapes
- The principles include balancing the elements of the design and having good contrast between various elements

Voice User Interfaces (VUIs)



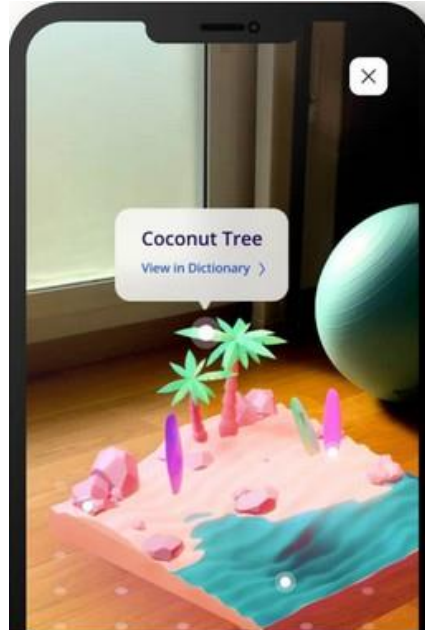
- Intuitive and natural way that users can use to interact with different mobile applications
- This is especially beneficial for people with visual impairments and other disabilities as well as for those with limitations because of their environment

Bottom Navigation



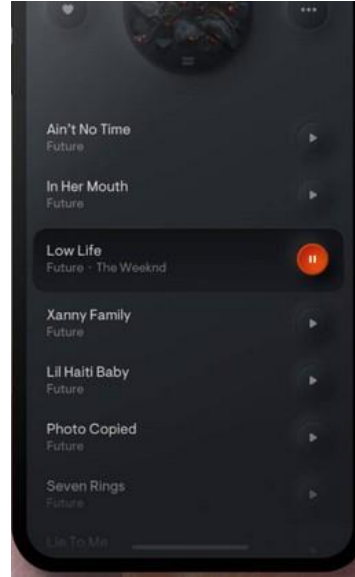
- When features are present at the screen's bottom, it becomes easier to access by thumb single-handedly
- Can be used to display around 4 to 5 destinations without the need to navigate through multiple options
- The features or destinations are labelled using either text or icons representing them

Augmented Reality



- This allows users to interact with virtual objects in a physical space
- Augmented reality is the new-age tool that seamlessly bridges the gap between the physical and the digital world
- Can help users visualise how the products will fit or look on them before making the final decision. This reduces returns and exchanges and increases sales of the products

Dark Mode



- Useful for reducing strain on the eyes and for improving battery performance
- Crucial for optimizing the user experience in low-light environments
- Reduces distractions and improves concentration
- Reading and editing on mobile devices becomes less strenuous and more interesting

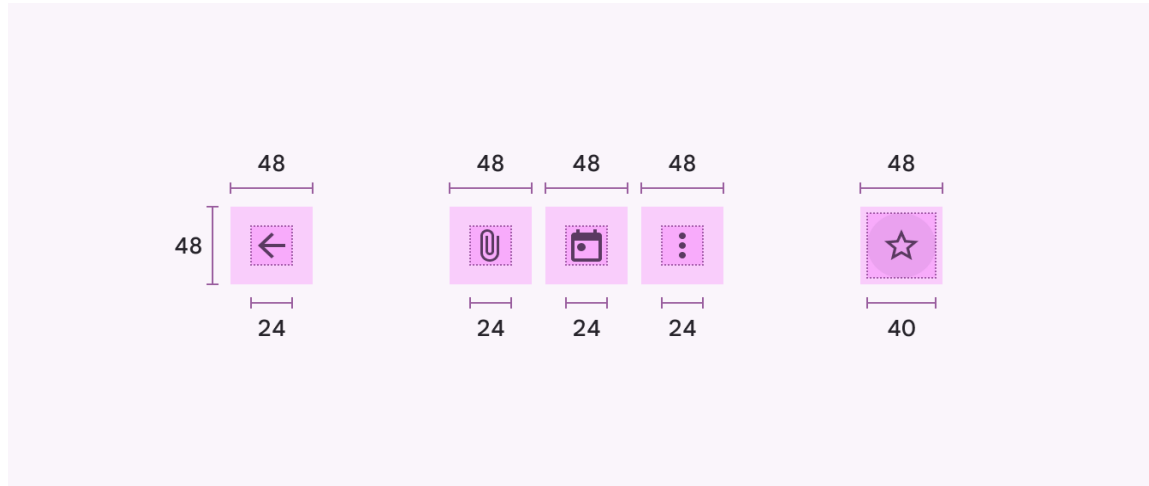
Glassmorphism



- The semi-transparent or fully-transparent backgrounds with blurred glass-like effects used in the current displays are popular as Glassmorphism
- The visuals of this mode are absolutely intriguing as it gives a sense of dimensionality and depth, making the app look edgy and visually classy
- The glass effects of this mode ensure transparency and openness, building credibility among users

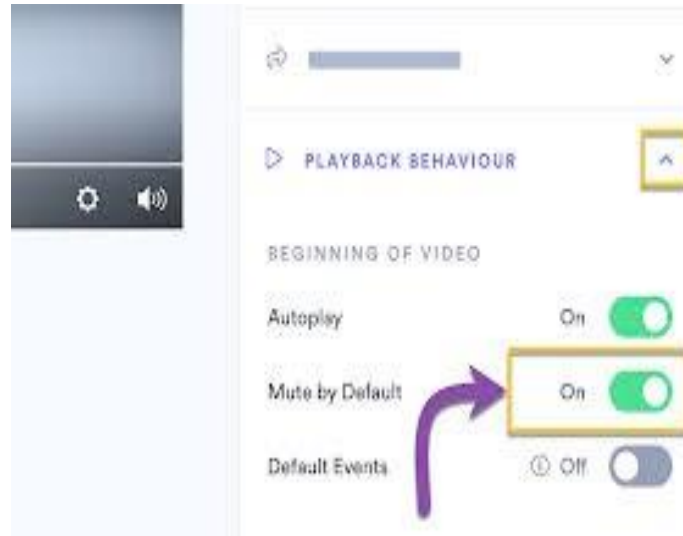
Common Mistakes in UX - Trends

Small Touch Targets on Touchscreens



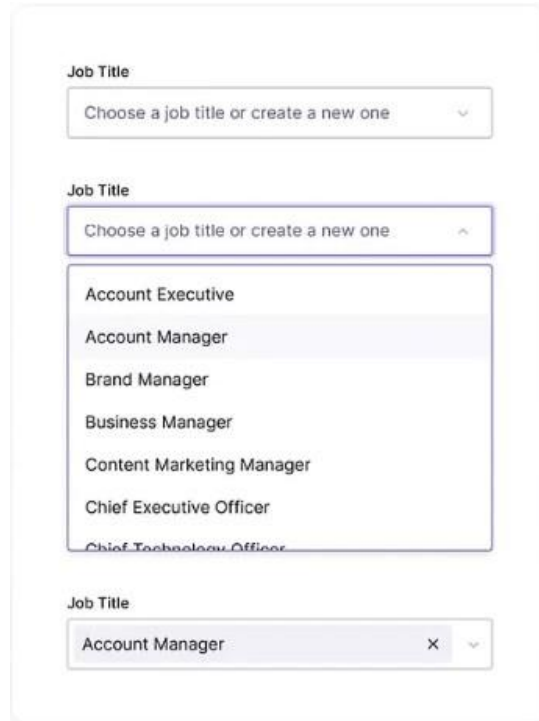
- Interactive elements must be at least 1 cm × 1 cm (0.4in × 0.4in) to support adequate selection time and prevent fat-finger errors
- An element like an icon may appear to be 24x24dp but the padding surrounding it comprises the full 48x48dp touch target
- Touch targets at least 48x48dp, separated by 8dp of space or more, to ensure balanced information density and usability
- Touch target of 48x48dp results in a physical size of about 9mm, regardless of screen size. The recommended target size for touchscreen objects is 7-10mm

Autoplay of audio/video



- One of the most annoying things that can overtake website users is the instantaneous audio playback when opening a web or mobile page
- Instead of forcing your website visitors to turn down the volume in a hurry, just provide a default mute with an option to turn it off

Long dropdowns



Job Title

Choose a job title or create a new one

Job Title

Choose a job title or create a new one

- Account Executive
- Account Manager
- Brand Manager
- Business Manager
- Content Marketing Manager
- Chief Executive Officer
- Chief Technology Officer

Job Title

Account Manager

- Have you ever tried to get to the needed item in a dropdown menu when it has to be scrolled?
- Believe us, this is not an easy task, especially for mobile users. Do not complicate their lives, this is not an easy thing

Violation of the three-click rule



- Have you checked how many clicks a user needs to make on your mobile app to complete the desired action?
- If for basic tasks this number exceeds three, something in your UX is definitely not right
- Although not been supported by data in any published studies to date (false rule), it is important to limit interaction cost

Checklists with a large number of options

Manage who can view Senior Living

×

Choose who can access this fund.

☐ All investors

☒ Only selected investors

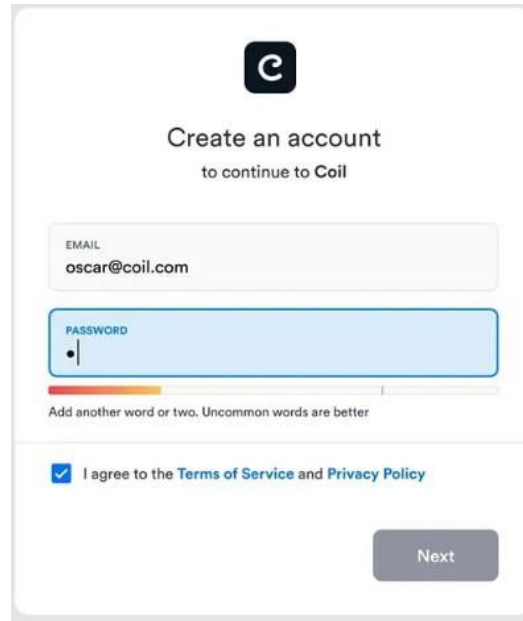
Selected investors who have access to this fund: [Select all](#) [Deselect all](#)

Name ▼	Grant access
PFA Pension	☒
AMF Pension Fund	☐
UBS AG	☒
VBL	☒
Deutsche Bank	☐
Stichting Pensioenfonds ARD	☒

Cancel

Save changes

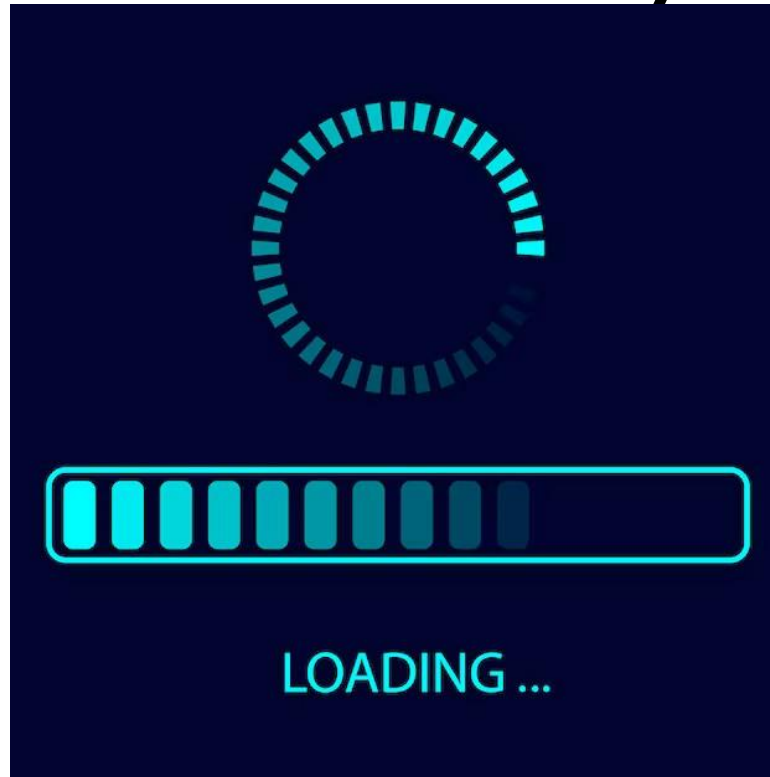
Complex password requirements



The screenshot shows a mobile app interface for creating an account. At the top is a dark square icon with a white 'c'. Below it, the text reads 'Create an account to continue to Coil'. There are two input fields: 'EMAIL' with the value 'oscar@coil.com' and 'PASSWORD'. The password field is highlighted with a blue border and contains a single character, with a strength indicator bar below it showing a gradient from red to yellow. Below the password field, a message says 'Add another word or two. Uncommon words are better'. At the bottom, there is a checked checkbox next to the text 'I agree to the Terms of Service and Privacy Policy', and a 'Next' button.

- Keep a balance between safety and usability
- Why allow only people with outstanding memory to access its resources?
- Apparently, only they will be able to access their accounts on your app due to the high password requirements

Load delay



- A few years ago, an acceptable download speed was equal to three seconds
- Today this number is much less

Additional Material

<https://design.google/resources/>

[Native-mobile-app-design-overall-principles-and-common-patterns-2017](#)

[UXpin/uxpin_mobile_ui_design_patterns_2014.pdf](#)

[12-mobile-ux-design-trends-for-2018](#)

<https://uxdesign.cc/ux-trends-2019-3ddadf853ca>

<https://www.mindinventory.com/blog/mobile-app-ui-ux-design-trends-2020/>

<https://www.designstudiouiux.com/blog/mobile-app-ui-ux-design-trends/>